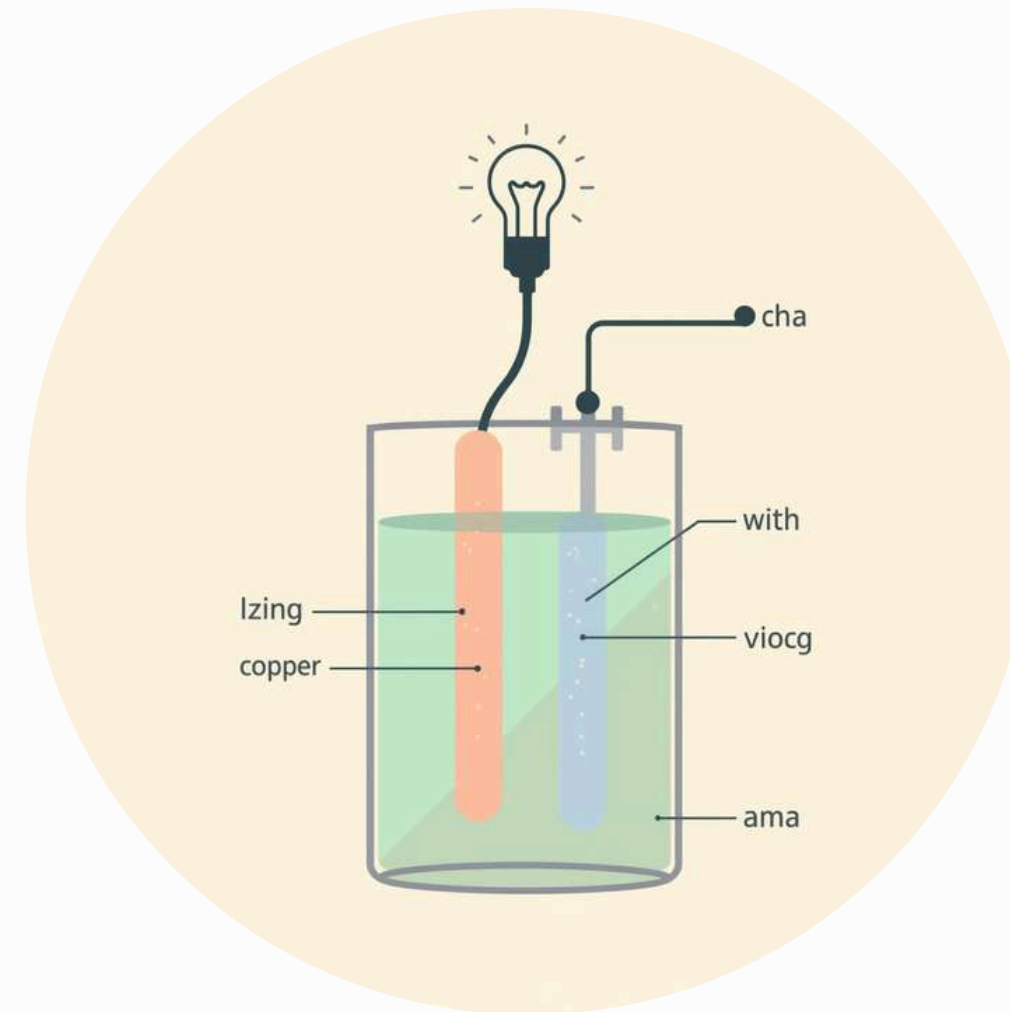


What is Electrochemistry?

Electrochemistry is how chemical reactions create electricity! It's the amazing science where chemistry and electricity work together.

Think about the batteries in your toys and remote controls — they all use electrochemistry to power up and work!





Meet the Galvanic Cell!

A galvanic cell is a tiny battery made by joining two different metals and a liquid that helps electricity flow. It's like magic — chemistry creates electricity!

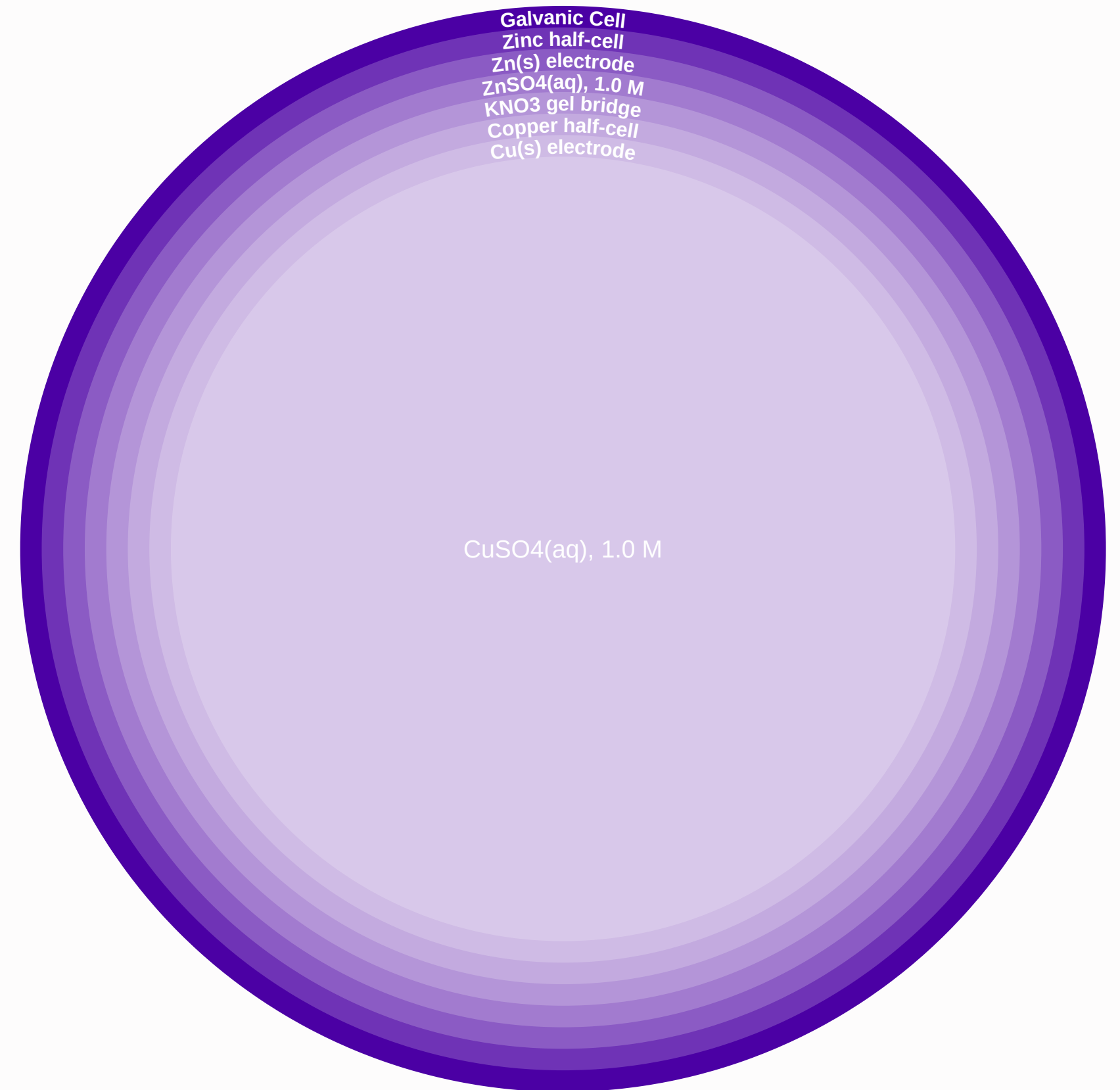
How Does a Galvanic Cell Work? (Part 1)

Think of it like a water slide for electrons! Electrons love to move from one metal to another. When we connect two different metals, electrons slide from the zinc to the copper, creating an electric current!

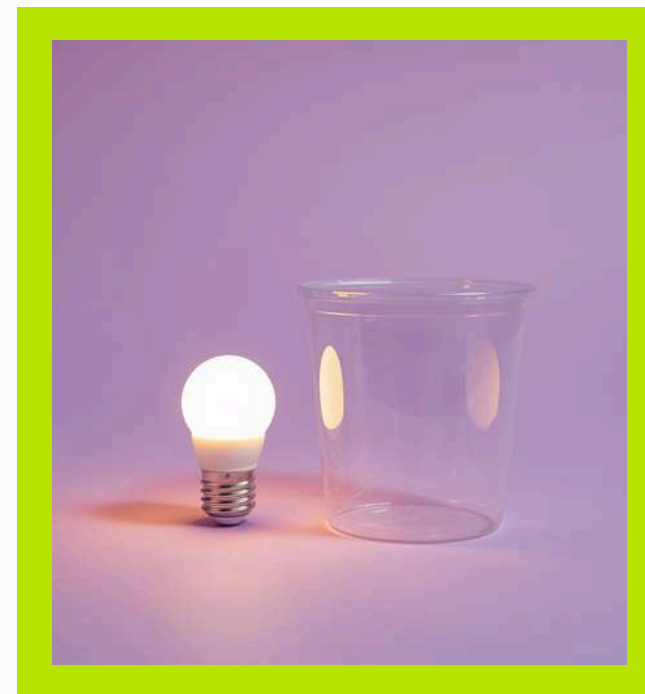


How Does a Galvanic Work? Cell

This diagram shows how electrons flow from the anode (zinc) to the cathode (copper) through the wire. The salt bridge allows ions to move between solutions, completing the circuit. This flow of electrons is what creates electricity!



- Materials You Need





Step 1:

Prepare Your Electrolyte

An electrolyte is a special liquid that helps electricity flow between the metals. You can make one easily at home!

Mix 2 tablespoons of salt into a cup of warm water and stir until dissolved. Or simply use fresh lemon juice as your electrolyte!



Step 2: Set Up Electrodes

Place the copper strip on one side of your container and the zinc strip on the other side. Make sure they don't touch each other! Leave some space between them in the electrolyte solution.

Step 3:

Connect the Wires to Electrodes

Attach one wire to the copper strip and another wire to the zinc strip using clips. Connect the other ends to your LED bulb. Make sure clips are secure and tight! The wires let electrons travel and flow.



Step 4:

Add the Salt Bridge

Soak a paper towel in saltwater and place it between your two containers or drape it over the edges to connect both solutions. This salt bridge allows ions to flow between the solutions, completing the circuit so electricity can flow!





Step 5: Watch It Work!

Look at your LED — it's glowing! Your galvanic cell is making electricity. Write down what you see. You did it — celebrate your success!

Safety First!

Stay Safe Always



Wear Safety Gloves

Always wear gloves when handling metals and solutions to protect your hands.



Adult Supervision

Work with a teacher or parent watching over you during the experiment.



No Tasting!

Never taste or drink solutions. Wash your hands thoroughly after the experiment.

Real-World Uses of Galvanic Cells

Galvanic cells power so many things we use every day! From the batteries in your toys and TV remotes to flashlights and clocks.

Even electric cars use special galvanic cells called rechargeable batteries to run their motors. Science is everywhere around us!



Recap: What We Learned Today



Chemical Reactions Create

Electricity Electrochemistry shows us how chemical reactions between metals can produce electrical energy we can use!

Galvanic Cells Are Tiny Batteries

We built our own battery using copper, zinc, and an electrolyte solution. Electrons flow from one metal to another!

Safety Is Important

Always wear gloves, work with adults, use safe materials, and wash hands after experiments.

Let's Test Your Knowledge! Quiz Time

Answer these questions:

- What two metals did we use in our cell?
- What is the role of the electrolyte?
- Why do we need a salt bridge?

Discuss with your partner!





Thank You & Keep for Exploring!

Great job building your galvanic cell!
Keep exploring science every day!



You're amazing scientists!
Next lesson: More electricity experiments!
Stay curious and keep learning!

